ARISE Bench — *Automatic Relational Inference and Structure Evaluation*

Team: Leo Yuhao Jin, Vivian Peng

Problem Statement

The way we evaluate learning in frontier AI systems is changing. Early benchmarks such as **MMLU** and **BIG-bench** mainly measured broad knowledge and one-shot reasoning. But there is still a major gap: most existing benchmarks do not directly measure **how a model learns**. Many tasks are still effectively static or single-shot, which makes it hard to tell whether a model is truly learning new structure or simply matching patterns it has seen before. They also tend to collapse different abilities into a single score, mixing together structure induction, belief updating, and compositional generalization. In addition, most benchmarks are not grounded in matched human behavioral data, making it difficult to compare model learning strategies with human learning.

We introduce **ARISE Bench**, a compact, human-grounded benchmark for **self-driven relational learning**. ARISE Bench is built from two tasks: **Transitive Inference** and **Learnability**. Together, these tasks measure key components of learning that has not been focused on before: How the model infer whether the environment is worth learning (learnable) by acquiring latent structure from sparse evidence and updating beliefs after corrective feedback. ARISE Bench provides a more reliable and more diagnostic way to evaluate adaptive learning in frontier models.

Task & benchmark construction

Task 1: Transitive Inference

Benchmark:

https://www.kaggle.com/code/vivpeng/benchmark-task-transitive-inference/edit/run/311069735

Transitive inference (TI) tests whether a model can infer the relative order of two items it has never directly compared by chaining through intermediate relations it has seen before. In the classic paradigm, subjects are trained only on adjacent pairs from a hidden linear hierarchy—for example, **A > B, B > C, C > D, D > E, E > F**—and are then tested on novel non-adjacent pairs such as **B > D**. Solving these held-out pairs requires an internal representation of the latent order rather than simple memory for observed comparisons.

Task 2: Learnability

Benchmark:

https://www.kaggle.com/code/vivpeng/benchmark-task-learnability/edit/run/311553852

The learnability task tests whether a model can detect when one stimulus set contains a more coherent latent structure than another, and whether this difference appears in its inferred ordering behavior rather than only in overall accuracy. The goal is not simply to ask whether the model can learn from feedback, but whether it shows stronger evidence of **deep relational organization** when the underlying structure is more learnable.

Technical Details

Task 1: Transitive Inference (TI)

This task tests whether a model can infer the correct order between two items it has **never directly seen compared**.

Setup

- Each episode contains a hidden 5-item ranking:
  - `x1 > x2 > x3 > x4 > x5`
- The model only sees **adjacent-pair evidence**:
  - `x1>x2`, `x2>x3`, `x3>x4`, `x4>x5`
- The final question is a **held-out non-adjacent pair**

What varies

- **Symbolic distance** of the held-out pair: `2, 3, 4`
- **Evidence budget**: `4, 8, 12` training observations

Primary metric

- **Held-out non-adjacent accuracy**

Diagnostic metrics

- Accuracy by symbolic distance
- **SDE slope:** the slope of accuracy as a function of symbolic distance (2–4), where a positive slope indicates that farther-apart pairs are easier
- **TIE = terminal accuracy − interior accuracy:** a measure of whether pairs containing an endpoint item are easier than middle-item pairs

Why it matters

A model that really learns the latent order should do well on unseen non-adjacent pairs, especially as distance increases. This helps separate true relational inference from shallow memorization.

Task 2: Learnability (Reward Yoking Version)

This task tests whether a model can tell the difference between a set with **real latent structure** and a set where feedback does **not support a stable ordering**.

Setup

There are two 5-item sets per episode:

- **Learnable set (L)**:
  - `x1 > x2 > x3 > x4 > x5` trained on adjacent pairs `x1>x2`, `x2>x3`, `x3>x4`, `x4>x5`
  - Reward follows the true latent order
- **Unlearnable / comparison set (U)**:
  - No reliable underlying order `x1 >/< x2 >/< x3 >/< x4 >/< x5`
  - Reward is randomly assigned independently of choice. However, reward probability is matched to recent performance on L. This ensures that the benchmark is testing sensitivity to **structure**, not just sensitivity to how often reward occurs.
- L and U trials are **interleaved**
- Test includes **all within-set pairs**, including non-adjacent ones
- The prompt includes recent trial history: items, choices, and rewards

Primary metric: Ordering Preference (OP) in both sets

For all within-set pairs (including non-adjacent ones), for each item, we compute its choice frequency during test trials:

```
choose_freq[item] = times item chosen / times item appeared
```

We then sort the frequency from highest to lowest and fit a linear regression

The slope β_1 is the **ordering preference (OP) score**:

- a **more negative slope** means the model exhibited strong ordering behaviors
- a slope near **zero** means weak or absent ordering preference

We then denote:

- **OP_L**: ordering preference for the learnable set
- **OP_U**: ordering preference for the unlearnable set
- **OP_discrimination = OP_L- OP_U**

Therefore, **to successfully detect the learnability/differentiate learnable from unlearnable environments**, it should show:

- a clear ordering preference on **L (<-0.1)**
- little or no ordering preference on **U (around 0)**
- Overall high level of OP_discrimination **(<-0.15)**

Why it matters

The learnability benchmark tests whether an AI model can detect that some stimulus sets have consistent learnable structure while others do not — and allocate its learning accordingly. This goes beyond measuring raw accuracy: a model that develops strong ordering preferences for learnable items but not for reward-matched unlearnable ones is tracking the statistical regularity of feedback, not just its magnitude. The design is adapted directly from monkey dACC electrophysiology experiments where learnability is encoded as a neural signal distinct from reward value [1, 2, 3]. A model that passes this benchmark is demonstrating a capacity central to general intelligence — knowing not just *how* to learn, but *when* learning is worth doing.

Dataset

Besides all three task has objective metric evaluating the performance, we also collected human dataset for both tasks, which provide several important insights that shaped both the design of the task and the evaluation metrics.

Transitive inference

Human dataset. Collected from 52 participants in a lab-based 5-item TI task. Each participant completed ~244 trials.

AI dataset. Generated by running each model on 20 independent sessions with randomized item labels.

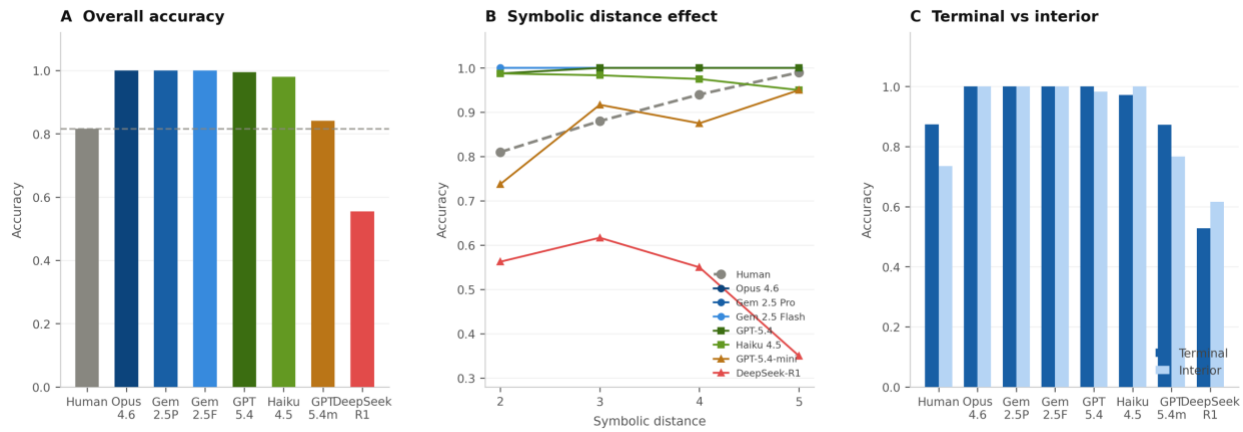
Learnability

Human dataset. Collected from 82 participants in a lab-based 5-item TI task, Each participant completed ~488 trials

AI dataset. Generated by running each model on a single session per seed with randomized stimulus labels.

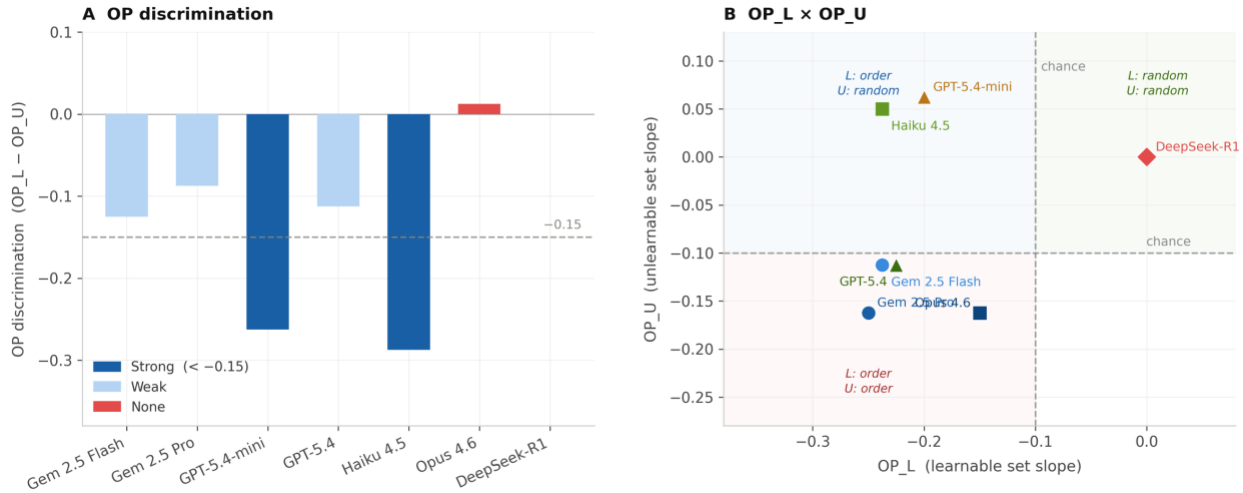
Results, insights, and conclusions

Transitive Inference

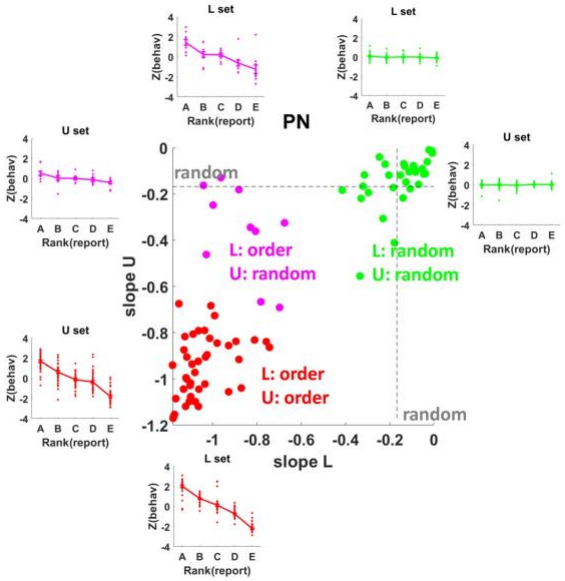


The TI task revealed a sharp capability split across models. Three frontier models — Claude Opus 4.6, Gemini 2.5 Pro, and Gemini 2.5 Flash — hit the ceiling at 100% overall accuracy, well above the human mean of 81.6%. GPT-5.4 (99.5%) and Claude Haiku 4.5 (98.0%) also substantially exceeded humans. GPT-5.4-mini (84.1%) performed closest to the human level, and DeepSeek-R1 (55.5%) fell near chance. Critically, ceiling performance by frontier models collapsed the two classic human signatures: the symbolic distance effect (SDE) and the terminal item effect (TIE) were both absent in models with 100% accuracy, as there is no variance left to differentiate. GPT-5.4-mini showed a positive TIE (0.106) and a non-monotone distance profile resembling human behavior, suggesting its errors carry structural information. DeepSeek-R1 showed an anti-SDE pattern — accuracy actually declined at distance 5 — inconsistent with any form of transitive inference. These results confirm that despite overall most of the AI models can learn TI, there are variabilities of the minuscule level of learning strategies among them.

Learnability



The learnability task showed clear variation across models. In Panel B, the chance threshold at −0.10 divides the OP_L × OP_U space into four interpretable quadrants. The upper-left quadrant (OP_L below chance, OP_U above chance) represents **the ideal learnability signature — ordering preference for learnable items but not unlearnable ones**. Claude Haiku 4.5 and GPT-5.4-mini both fall in this quadrant, with OP_disc values of −0.288 and −0.263 respectively, reflecting the strongest discrimination between conditions. GPT-5.4 and Gemini 2.5 Flash also show OP_L below chance but with OP_U similarly below chance, landing in the lower-left quadrant (L: order, U: order) — exhibiting ordering preferences for both stimulus sets, consistent with a general tendency to impose structure rather than selectively tracking learnability. Gemini 2.5 Pro falls in the same quadrant with slightly weaker overall slopes. Claude Opus 4.6 sits near the boundary with OP_L marginally below chance and OP_U below chance as well, suggesting limited but non-selective ordering. DeepSeek-R1 remains at the origin (0, 0), showing no ordering preference for either stimulus set.



The pattern of individual variability in AI models echoes findings from the human PN data. In the human study, subjects naturally clustered into three groups: those who developed ordering preferences for both learnable and unlearnable sets (L: order, U: order), those who showed ordering only for the learnable set (L: order, U: random), and those who showed no consistent ordering for either (L: random, U: random). The fact that the same three-cluster structure observed in human subjects under the PN schedule reappears across AI models — without any deliberate design — suggests these behavioral profiles may reflect fundamental differences in how systems track and exploit statistical regularity in feedback, rather than being unique to biological learning.

Organizational affiliations

Vivian Peng — Product-minded software engineer; builds efficient and clean AI benchmark code and model evaluation

Leo Yuhao Jin — PhD candidate (Columbia, Neurobiology); leads idea conceiving, dataset collection and overall management

References & citations

- Jin, Y., Jensen, G., Gottlieb, J., & Ferrera, V. (2022). Superstitious learning of abstract order from random reinforcement. *Proceedings of the National Academy of Sciences*, *119*(35), e2202789119. https://doi.org/10.1073/pnas.2202789119
- Jin, Y., Jensen, G., Ferrera, V., & Gottlieb, J. (2025). Single-neuron encoding of learnability in the dorsal anterior cingulate cortex. *bioRxiv*. https://doi.org/10.1101/2025.09.29.679390
- Jin, Y., Jensen, G., Gottlieb, J., & Ferrera, V. (n.d.). Learning of predictable rules mixed with random reinforcement in humans and neural networks. *Manuscript in preparation.*